



PROFESSIONAL SUMMARY

Mechanical engineering graduate with practical design and testing experience, including a successfully completed capstone project. Proficient in SolidWorks and MATLAB. Passionate about UAV systems and looking to contribute to a team building the next generation of drone transportation technology.

EDUCATION

San Diego State University | San Diego, CA

Bachelor of Science in Mechanical Engineering (GPA: 3.6, Dean's List: Fall 2024)

Expected Graduation May 2026

Masters of Science in Mechanical Engineering

Expected Graduation May 2028

ENGINEERING EXPERIENCE

Senior Design Capstone Project – C-UAS Competition Team | Team Lead | SDSU

September 2025 – Present

- Collaborated with 5 Mechanical Engineering, 4 Aerospace Engineering and 5 Electrical and Computer Engineering (14) teammates to design and manufacture an autonomous Bi-Plane Tail-Sitter VTOL over a 10-month development cycle resulting in an air worthy and competition ready drone.
- Designed a master assembly containing 30+ unique parts in SolidWorks using DFA principles, performed virtual fitment and interference analysis across all mechanical subsystems to ensure 100% tolerance compliance before physical assembly.
- Conducted motor characterization testing across full throttle range, generating thrust-vs-current efficiency curves that identified a 17% thrust deficit in a propulsion unit, directly informing a V1-to-V2 platform redesign that scaled the drone size by 33%.
- Developed and executed dimensional tolerance testing protocols using FDM additive manufacturing, iterating on critical fit interfaces to ensure mechanical compatibility across subsystems prior to full assembly.
- Engineered an autonomous VTOL UAS by applying DFM and DFA principles for FDM additive manufacturing, optimizing part orientation and support structures to increase part strength by 12% while reducing print time by 7% ensuring structural integrity for mission-critical flight.
- Produced and maintained test documentation across motor, structural, and flight test phases, creating a traceable record of design decisions that supported faculty reviews and contributed to securing \$2,000 in additional project funding

BRDG Innovation Challenge | Hardware Team Lead (1st Place Winner)

June 2025 – October 2025

- Collaborated on a multi-disciplinary team to design and prototype an interactive smart recycling bin with the goal of educating the K-12 students on sustainable waste management.
- Designed a servo-actuated smart recycling bin in SolidWorks with PLA 3D-printed linkage arms and plywood enclosure; delivered a functional prototype on budget.
- Diagnosed and resolved latency issues in detection and actuation pipeline by migrating to an optimized Python library and refactoring control logic, achieving a 43% improvement in response time and detection accuracy.
- Won \$3,500 first-place prize competing against multidisciplinary student teams, received further support from the organizers to co-fund a startup with the team.

WORK EXPERIENCE

Manager | Cold Stone Creamery La Jolla

January 2018 – January 2026

- Supervised and trained 20+ employees; directed daily operations to maintain high customer satisfaction during high-volume shifts.

STEM Facilitator | Treobytes, San Diego

July 2024 – August 2024

- Developed and led engaging STEM content for a summer camp for 15-30 middle school students including students with learning disabilities, achieved 100% program completion rate.

CAMPUS INVOLVEMENT

Co-founder/Treasurer | San Diego State University nthTry Engineering Club

September 2024 – September 2025

TECHNICAL SKILLS

SolidWorks(Advanced) * Matlab(Intermediate) * ANSYS(Intermediate) * Python(Intermediate) * Excel(Intermediate) * GD&T
DFA * DFM